

February 6, 2026

To Whom It May Concern,

I am Sinchan Banerjee, a Director of Enterprise Data Architecture at UST, where I serve clients across the healthcare sector. Over the past 15 years, I've led initiatives in enterprise data platforms and digital transformation. I specialize in architecting scalable highly complex high volume data engineering solutions and emerging AI workloads. I've pursued executive education from MIT (Artificial Intelligence in Healthcare), Cornell University (Executive Leadership Program), and UC San Diego (Machine Learning). I earned my Bachelor of Technology in Computer Science Engineering from SRM University. My technical contributions include a patent on predictive analytics and authorship of widely accepted book on data engineering - "Scalable Data Architecture with Java" (Packt Publication, 2022). I've published many research on various international journals, presented at premier conferences and served as a judge and reviewer in multiple STEM and technology-related initiatives. I also contribute to the technical community as a Senior IEEE member. My expertise and seniority makes me an authority to judge and comment on Vivek's exceptional abilities.

I first met Vivek in December 2024 at an invited industry-academia hybrid lecture series hosted by Anand International College of Engineering, where we both spoke in the same session <https://anandice.ac.in/latest-news/lecture-series>. In that event, I presented my work on cloud-native data platforms and real-time healthcare analytics. At the same time, Vivek delivered a talk on his applied research in artificial intelligence, intelligent transportation systems, and the backend systems that support real-world infrastructure decision-making. From that first interaction, it was clear that he operates comfortably at the intersection of advanced AI research and practical software engineering, with a focus on systems that directly support public-sector and infrastructure-related decisions.¹

We met again in March 2025 at AICTE-approved Faculty Development Program. AICTE (All India Council for Technical Education) is India's national regulatory body for technical and management education, responsible for planning, coordinating, and ensuring quality standards in engineering, technology, management, pharmacy, and architecture fields organized by the same college, where we were once more co-speakers in a session focused on emerging technologies for

¹ First India news report on the "Bridging Industry & Academia" lecture series (December 8, 2024): firstindia.co.in/article-view/jaipur/08122024/20987

LinkedIn post from Anand International College of Engineering mentioning the event: LinkedIn Post. https://www.linkedin.com/posts/anand-college-of-engineering_anandice-industryknowledge-hybridlectureseries-activity-7265909167196954624-uyYe?utm_source=share&utm_medium=member_desktop with Vivek's present on picture. Dainik Bhaskar news report (December 14, 2024): bhaskar.com/epaper/detail-page/jaipur/14/2024-12-14?pid=13

engineering education and public systems. In that program, Vivek gave a particularly impactful presentation on how large language models and agent-based simulations can support planning and policy in intelligent transport systems². He explained how AI can be integrated with public sector infrastructure modeling to examine transportation equity, network performance, and policy trade-offs, referencing research he conducted at California State University, Long Beach, that international researchers and transportation authorities have already cited.

In our follow-up conversations during and after that event, he also described his experience as a backend engineer responsible for scalable service infrastructure and secure system integration in enterprise environments, which reinforced my impression that he is not only a theorist but someone who understands how to turn ideas into robust, real-world systems.

In my view, Vivek's most significant contributions center on bringing AI, simulation, and backend engineering together to address real-world infrastructure and policy challenges. This assessment is grounded not only in his research and presentations but also in his hands-on experience as a backend engineer responsible for crucial components in scalable enterprise environments.

His work on the Chase Shopping Hub project at JPMorgan Chase, for example, involved building an API from scratch and integrating essential services that directly impact the experience of millions of customers, demonstrating his ability to translate concepts into robust, real-world systems. Furthermore, the relevance of his research in intelligent transportation systems and simulation is confirmed by the fact that his contributions have already been cited in government research projects in places like Florida, Italy, Iran, and Colombia, indicating a significant impact that extends beyond his immediate academic circle.

Rather than treating AI as an abstract optimization problem, he uses it to explore how different groups and regions are affected by transportation policies, and how systems can be designed to be more inclusive and transparent. At the same time, his solid background in backend engineering allows him to bridge the gap between research prototypes and production-ready systems, ensuring that the tools and models he works on can be integrated into larger enterprise and public-sector environments. This dual strength, deep AI modeling combined with hands-on backend infrastructure expertise, is what, in my opinion, makes his profile distinctive in both research and practice.

The significance of these contributions is substantial in the current technology and public-policy landscape. Governments and transportation agencies around the world are seeking data-driven and

²LinkedIn post from Anand International College of Engineering mentioning the event: LinkedIn Post https://www.linkedin.com/posts/anand-college-of-engineering_ai-machinelearning-fdp2025-activity-7316862944720367616-6E2J?utm_source=share&utm_medium=member_desktop&rcm=ACoAAA5MBBEbT2_lbHsfj2eSfXiW5lqXKeCrgaM

AI-supported methods to plan infrastructure, evaluate equity, and justify investments, and Vivek's work directly supports these needs. His AI-focused research has already been cited by public-sector researchers across countries, specifically in government research projects in Florida, Italy, Iran, and Colombia, indicating that his ideas and tools are being used beyond his immediate academic circle.

The ability to apply machine learning and simulation to transportation policy and planning in ways that account for equity, explainability, and real-world constraints is both timely and challenging to find. On the engineering side, his experience building backend systems that integrate with enterprise platforms brings a hands-on practicality that complements his theoretical expertise, making it much more likely that his innovations can survive beyond the lab and be deployed in production environments where reliability, security, and scalability matter.

Based on our subsequent, in-depth discussions about his professional trajectory and the comprehensive materials he shared with me, I have a clear understanding that Vivek has also held leading and critical roles in his academic and professional collaborations. In his work with California State University, Long Beach, and the California Department of Transportation, he contributed essential technical expertise in data analysis, simulation, and system modeling to projects that support transportation research and policy.

He played a key role in implementing Python-based solutions and AI frameworks that underpinned research outputs later cited by scholars and agencies worldwide, which is a strong indicator that his contributions were both central and non-trivial. In his industry engagements, while his formal role titles may be more technical than managerial, he has taken ownership of key backend components and mentored his team, demonstrating leadership through accountability, technical guidance, and knowledge sharing rather than through formal hierarchy.

Another important marker of his standing in the field is his being trusted to judge and evaluate others' work. Vivek has served as a judge at hackathons and has mentored students and professionals working in AI, transportation, and healthcare technology, helping shape early-stage projects into more rigorous and impactful solutions.

He has also reviewed papers for respected conferences such as NeurIPS and ICLR, and has served as an evaluator for IEEE Senior Member applications, roles generally offered to individuals whose expertise and judgment are respected by the broader technical and academic community. In addition, his contributions to venues such as Informatica.si show that he continues to participate in peer-reviewed and public-facing research efforts, further reinforcing the recognition of his expertise.

Given the standards of U.S. immigration regulations for the EB1-A category - specifically, the requirement that the beneficiary be among the small percentage who have risen to the very top of the field and who have sustained national or international acclaim - it is my professional opinion

that Vivek fits well within this framework. He is an emerging leader in AI-driven infrastructure systems and backend engineering, with research in intelligent transportation and simulation that has been cited internationally and technical contributions to enterprise-grade systems that demonstrate depth, reliability, and impact.

What sets him apart is the consistency with which he operates across research, engineering, and public-good applications: he works on problems that matter to governments and agencies, builds deployable systems, and actively participates in the community by judging, reviewing, and mentoring.

From my vantage point as a senior professional in enterprise data and AI systems for healthcare, I see Vivek as a technologist whose work is not only innovative but also socially relevant and scalable. He operates in domains critical to the future of AI in infrastructure and public systems, and his record of technical contributions, research recognition, and evaluative roles demonstrates that he is far more than a competent engineer.

He is helping define how AI and backend systems are applied to real-world challenges in transportation and beyond. For these reasons, I view his profile as aligned with the EB1-A category's expectations: his contributions are original, impactful, and increasingly global in scope, and I strongly support his immigrant petition to the United States.

Best Regards,

A handwritten signature in black ink, appearing to read 'Sinchan Banerjee', with a stylized, cursive script.

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